
LabFab: A custom lab electrode fabricator

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ABSTRACT This work presents the design and validation of a prototype platform for automated electrode fabrication, consolidating five sequential steps—surface roughening, slurry preparation, blade coating, drying, and adhesion testing—into a single workflow. Profilometry confirmed that coarser grit treatments enhanced copper surface roughness, with Ra values of 1.87 μm (120/240 grit), 1.26 μm (240/400 grit), and 1.02 μm (400/800 grit), where the higher roughness provided better anchoring sites for active material deposition. Slurry viscosity was maintained in the range of 7.6–10.4 mPa-s, ensuring uniform blade-coating performance, while soak tests demonstrated >90% adhesion stability after 12 hours of electrolyte exposure. Unlike conventional manual fabrication, our platform integrates all steps into one automated process, thereby improving reproducibility and reducing operator variability. By accelerating electrode prototyping, this system enables rapid material screening and supports the development of next-generation batteries and supercapacitors. Future work involves conducting extensive electrochemical tests to determine the performance improvement over manually coated test electrodes.

I. INTRODUCTION

The development of advanced energy storage solutions, ranging from high-performance batteries to efficient supercapacitors, critically relies on precise electrode fabrication. At the heart of this process lies the preparation of electrode slurries, where rheological properties are paramount. Slurry viscosity, in particular, plays a fundamental role in determining the quality and uniformity of the electrode coating, directly influencing the device's performance and manufacturability [1], [8], [9], [10], [13], [14]. Extensive studies have shown that parameters such as solid content, binder type, and mixing conditions significantly impact the rheological behavior, which in turn affects coating stability and the resulting electrode microstructure [1], [4], [14], [22]. Constructing micro-rheological models for lithium-ion battery electrode slurries is crucial for predicting coating windows, enhancing process control, and understanding complex flow dynamics [12], [11].

Strategic decisions in large-scale automotive battery cell manufacturing are deeply intertwined with operational factors like slurry handling, drying rates, and calendaring pressures [2], [21]. These elements collectively influence manufacturing costs, throughput, and product consistency [2]. Among the post-processing steps, calendaring has emerged as a vital technique to enhance electrode density and improve electrochemical performance [5], [16]. The impact of calendaring on electrochemical performance has been extensively investigated, highlighting its importance for lithium-ion battery production [5], [16]. This necessitates detailed modeling and simulation, including discrete element method models and digital twin approaches, to optimize the calendaring process and other manufacturing steps [5], [27], [28], [29]. Such digital twin

platforms offer flexible and scalable solutions for optimizing the entire battery manufacturing chain [29], [28].

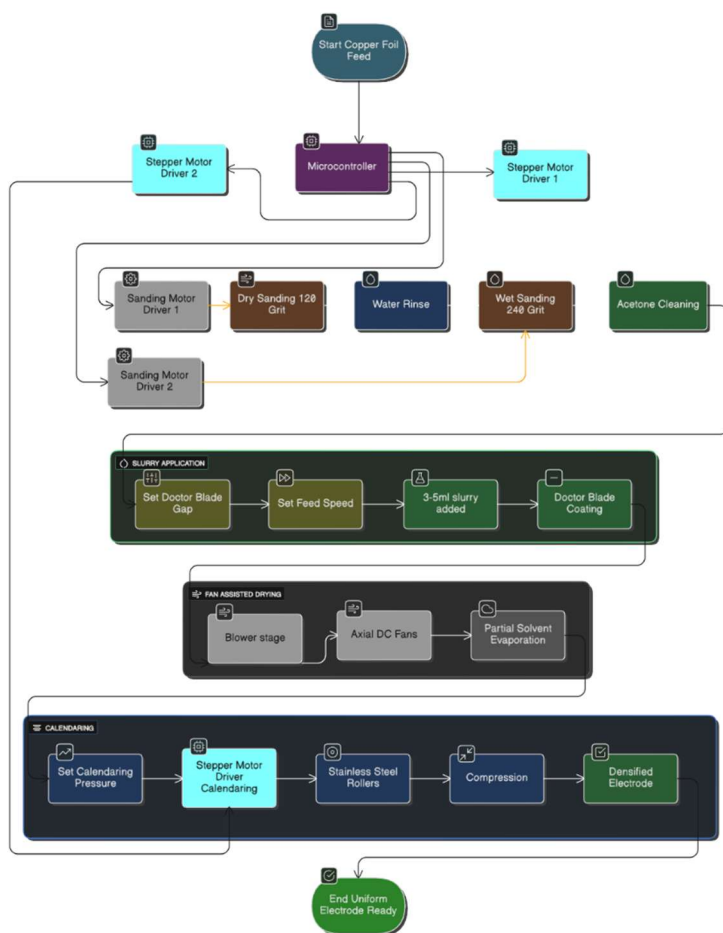
Our project focuses on LabFab, a custom-made, portable device designed for preparing electrodes across a wide array of energy storage applications. For all test innovations, our decision to utilize a PVA-based slurry aligns with advancements in binder chemistry. Novel binding agents and materials, including polyvinyl alcohol (PVA)-blocked polyurethane networks and other electrochemically active materials, have demonstrated improved mechanical and adhesive properties of electrodes while maintaining compatibility with active materials [3], [4], [17], [18], [23]. The influence of binder crystallinity on anode performance, for instance, underscores the importance of binder selection [17]. Furthermore, the development of water-soluble PVA as a functional binder for high-sulfur-loading cathodes represents a cost-effective approach to improving performance [18].

Ensuring the quality and consistency of electrode fabrication also involves advanced metrology and in-line quality control techniques [6]. These methods are gaining attention for their ability to monitor critical attributes such as coating thickness, surface roughness, and other crucial parameters in real-time [6]. Automated platforms for electrode formation and 3D printing techniques are also revolutionizing fabrication processes, enabling standardized graphite electrode creation and large-scale manufacturing of ion-selective electrodes for various diagnostic applications [19], [20]. The effect of blade-coating conditions on electrical and optical properties of nanowire electrodes further exemplifies the need for precise control [15]. Additionally, experimental and computational analyses of slurry-based manufacturing for solid-state battery

composite cathodes are advancing our understanding of these complex systems [7]. Research also extends to spray deposition techniques for supercapacitor electrodes using aqueous dispersions, showcasing the versatility of slurry-based methods [30].

Collectively, these advancements, from fundamental rheological studies [1], [8], [9], [10], [11], [12], [13], [14] to sophisticated manufacturing optimizations [2], [5], [6], [15], [16], [19], [20], [21] and the integration of digital twin technologies [27], [28], [29], contribute toward more reliable, scalable, and cost-effective electrode manufacturing processes [2], [25], [26]. This holistic approach facilitates the accelerated deployment of high-performance energy storage devices in critical sectors such as electric vehicles and portable electronics, with ongoing research continuing to refine every aspect of the battery electrode manufacturing roadmap [26]. Furthermore, advanced electrochemical models are crucial for revealing how manufacturing parameters affect battery performance, pushing the boundaries of what's possible [24].

II. Working



Substrate Sanding and Cleaning

Copper foil undergoes multi-step preparation for adhesion: dry sanding with 120-grit abrasive to remove oxides, wet sanding with 240-grit under water rinse for uniform texture, and acetone cleaning to eliminate organic contaminants. This ensures both mechanical roughness and chemical cleanliness.

Parameter	Specification	Value/Setting
Dry sanding abrasive	silicon carbide	120 grit
Dry, Wet sanding motor	DC motor speed	60 RPM
Wet sanding abrasive	silicon carbide	120 grit
Contact pressure	Sanding pressure	3 N (steel springs)
Cleaning solvent	Acetone	(99.9% pure)

Slurry Coating (Doctor Blade)

A moving copper strip is coated with slurry (carbon-based mix in ethanol) using a fixed doctor blade with micron-level gap adjustment (50–500 μm). Stepper motor-driven feed and calendaring rollers maintain uniform speed and tension, ensuring consistent coating thickness over long lengths.

Parameter	Specification	Value/Setting
Blade gap range	Micrometer screw	50–500 μm
Blade gap setting	Actual gap	~60 μm
Feed motor	NEMA 17 stepper	adjustable
Calendaring motor	NEMA 17 stepper	adjustable
Conveyor speed	linear velocity	15 mm/min
Slurry composition	Activated carbon + PVA + ethanol	-

Fan-Assisted Drying

Coated electrodes pass through a chamber with axial DC fans providing ambient airflow to enable partial solvent evaporation. This stabilizes the wet slurry film, reducing tackiness and smearing before compression, though full oven drying is still required later.

Parameter	Specification	Value/Setting
Fan type	Axial DC, 12V	3 A
Airflow velocity	Over coated surface	4.5 m/s dry

Calendering

The coated strip is compressed between stainless steel rollers under controlled pressure. This densifies the electrode, smooths the surface, improves adhesion, and reduces porosity, producing uniform electrodes ready for final drying and electrochemical testing.

Parameter	Specification	Value/Setting
Roller material	304 stainless steel	31 mm diameter
Roller pressure	Spring-loaded compression	0.9 kN/m
Roller gap	Adjustable roller spacing	Electrode thickness
Substrate tension	Controlled via drive system	~2 N

III. Characterization

A. Stylus Profilometry

Stylus profilometry was used to measure surface roughness (R_a), peak-to-valley height (R_z), and root mean square roughness (R_q) for substrates prepared using different grit combinations. The results show a consistent reduction in roughness with the use of finer grits, indicating improved surface adhesion for slurry coating. Table I summarizes the measurements obtained.

Grit Combination	R_a (μm)	R_z (μm)	R_q (μm)
120 and 240	1.87	12.6	2.40
240 and 400	1.26	10.0	1.64
400 and 800	1.02	8.97	1.25

Table I Surface Roughness Parameters for Different Grit Combinations

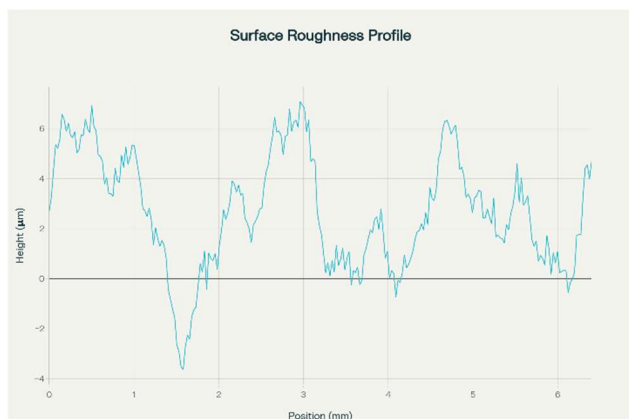


Fig. 1. Surface roughness profile obtained for copper sanded with 120 and 240 grit silicon carbide sandpaper using stylus profilometry at 0.5 mm/s over a 6.40 mm scan length as per ISO 1997.

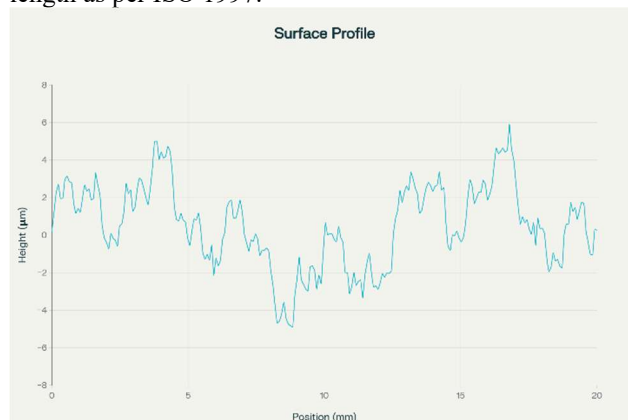


Fig. 2. Surface roughness profile obtained for copper sanded with 240 and 400 grit silicon carbide sandpaper using stylus profilometry at 0.5 mm/s over a 6.40 mm scan length as per ISO 1997.

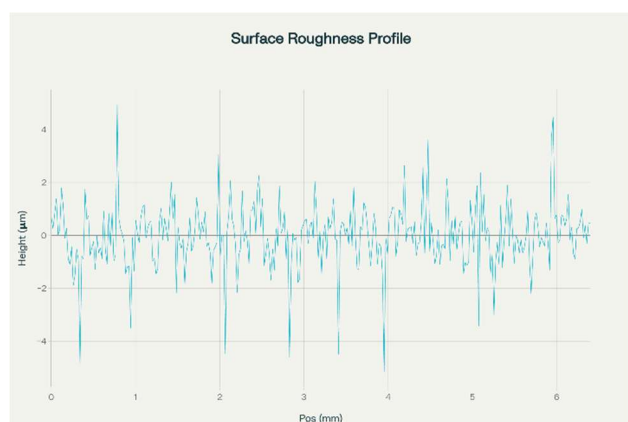


Fig. 3. Surface roughness profile obtained for copper sanded with 400 and 800 grit silicon carbide sandpaper using stylus profilometry at 0.5 mm/s over a 6.40 mm scan length as per ISO 1997.

B. Brookfield Viscometry

The viscosity of three dummy slurry samples was measured using a Brookfield viscometer, which provides accurate rotational viscosity readings in milliPascal-seconds ($\text{mPa}\cdot\text{s}$). Each sample consisted Graphite, polyvinyl alcohol (PVA) as the binder, and distilled water as the solvent. Viscosity was measured at room temperature under standard spindle and rotation speed settings. The observed increase in viscosity across samples is attributed to variations in binder concentration and dispersion quality. Higher PVA content contributes to increased network entanglement within the slurry, resulting in greater resistance to flow.

Slurry	PVA % (v/v)	Viscosity (mPa·s)
A	43%	7.6
B	50%	8.3
C	68.8%	10.4

Table II Brookfield Viscometry Results for slurry samples

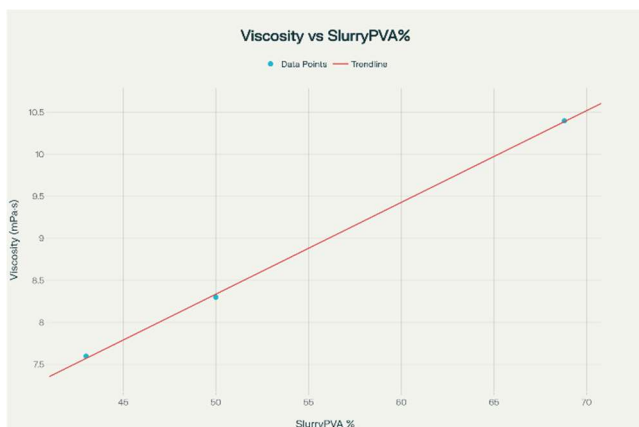


Fig. 1. Relationship between SlurryPVA% (v/v) and estimated viscosity (mPa·s). The graph presents experimental data points for three different SlurryPVA% values, with a linear trendline indicating a positive correlation between SlurryPVA concentration and viscosity.

C. Soak Test using 3M Na₂SO₄

A soak test was performed by immersing coated copper electrodes in 3 M Na₂SO₄ solution for 12 hours to evaluate adhesion under electrolyte exposure. Samples with higher PVA content and smoother substrates (400 and 800 grit) retained their coating integrity, showing no visible delamination or swelling. In contrast, samples with lower binder content or rougher surfaces exhibited partial peeling and softening. The test confirms that both binder concentration and substrate finish significantly affect the electrode's electrolyte stability.

D. Chronopotentiometry

Chronopotentiometry (CP) is an electrochemical technique where a constant current is applied to an electrode and the resulting voltage change is recorded as a function of time. This method enables the evaluation of electrode kinetics, charge storage mechanisms, and capacitive behaviour, making it an essential tool for optimizing supercapacitor electrodes.

In supercapacitor research, CP is commonly employed to assess specific capacitance, charge-discharge stability, and electrode performance. During a CP test, the electrode is subjected to a steady current, and the discharge voltage

profile is analysed. From the discharge curve, the specific capacitance (C_s) is calculated using the relation:

$$C_s = \frac{I \times \Delta t}{\Delta V \times m}$$

This method will be vital for the future optimisation and calibration of both the electrode coating machine as well as for deciding the slurry composition for a given blade gap.

D. Cyclic voltammetry

Cyclic voltammetry (CV) is an essential electrochemical technique used to characterize supercapacitor electrodes by sweeping the voltage between set limits and measuring the resulting current. This dynamic method provides valuable information on the redox behaviours, charge storage mechanisms, and electrochemical kinetics of electrode materials.

CV helps in identifying reversible redox peaks related to pseudo capacitance and distinguishes between capacitive and diffusion-controlled charge storage. By analysing the shape of the voltammograms at varying scan rates, it facilitates the evaluation of rate capabilities and double-layer capacitance characteristics. Furthermore, CV can monitor electrode stability by comparing current responses over multiple cycles.

E. Coating methods comparison.

Electrode coating can be performed manually or using the LABFAB electrode coating machine. Manual coating relies on operator skill and experience, often leading to non-uniform thickness, streaks, and variable edge coverage. These inconsistencies can adversely affect adhesion and electrochemical performance. In contrast, the LABFAB machine provides precise control over coating parameters, resulting in uniform thickness, smooth surfaces, and reproducible sample-to-sample quality. This reproducibility is critical when correlating substrate roughness, slurry viscosity, and adhesion properties. Therefore, LABFAB coating is preferred for controlled experimental studies and large-scale sample preparation.

Parameter	Manual Coating	LABFAB Coating
Coating Thickness	Variable; depends on operator	Uniform; controlled by machine
Surface Smoothness	Streaks, bubbles, uneven layers	Smooth and consistent
Edge Coverage	Often inconsistent	Precise and uniform

Parameter	Manual Coating	LABFAB Coating
Sample-to-Sample Reproducibility	Low; high variability	High; nearly identical coatings
Operator Dependency	High	Minimal
Scaling for Multiple Samples	Difficult	Easy; same settings for multiple electrodes

F. Cost and time analysis compared to commercial tools

The use of in-house electrode fabrication and characterization methods offers significant advantages in terms of cost and experimental flexibility compared to commercial laboratory tools. Manual coating and lab-prepared slurry systems are low-cost alternatives, requiring only raw materials (graphite, PVA, water) and basic instruments, with total consumable costs typically below ₹500–1000 per batch. The LABFAB coating machine, though an initial capital investment (~₹1–2 lakh), significantly reduces coating time per sample (from 10–15 minutes manually to 2–3 minutes per sample) and improves reproducibility.

In contrast, commercial coating and characterization systems are often highly expensive (₹5–10 lakh), require proprietary consumables, and involve additional maintenance and service costs. While commercial tools provide advanced features and automation, the in-house setup strikes a balance between cost-effectiveness, moderate throughput, and sufficient precision for research and small-scale experimentation.

Parameter	In-House Lab (Manual + LABFAB)	Commercial Lab Tools
Initial Cost	₹0–2 lakh (LABFAB only)	₹5–10 lakh
Consumable Cost per Batch	₹500–1000	₹2000–5000
Coating Time per Sample	10–15 min (manual) / 2–3 min (LABFAB)	1–2 min (automated, high throughput)

Parameter	In-House Lab (Manual + LABFAB)	Commercial Lab Tools
Reproducibility	Moderate (manual) / High (LABFAB)	Very High
Flexibility	High (adjustable slurry, volume, method)	Limited to machine settings

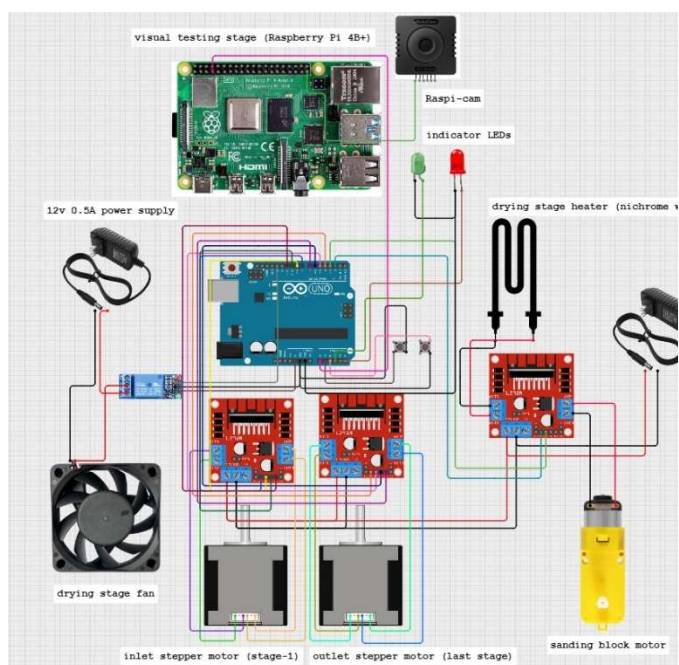
IV. Limitations and Strengths of the In-House Setup

The in-house electrode fabrication setup, including manual and LABFAB coating, offers several advantages but also has limitations. The main limitations include the lack of in-line drying temperature control, restricting optimization of binder evaporation and adhesion; compatibility limited to copper substrates; and testing restricted to carbon-based slurries, which may not directly extend to other active materials.

Despite these constraints, the setup exhibits several strengths. It is modular, allowing adjustments in slurry composition, coating thickness, and substrate handling. It is low-cost compared to commercial tools, yet provides high reproducibility, particularly when using the LABFAB machine. Additionally, the open-source design allows customization and scaling for experimental needs, making it highly suitable for research and development in energy storage applications.

V. Electronics used and circuit diagram

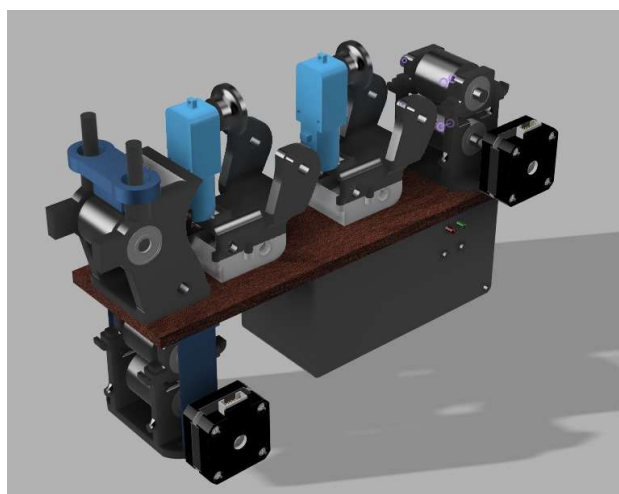
1. Stepper motor - NEMA-17HS4023
2. Motor driver - L298N
3. Arduino UNO R3
4. Fan - 12V 0.5A
5. DC motors - 3V to 9V



VI. Hardware used

1. Stainless steel annealed springs
2. 608ZZ ball bearings
3. 8mm diameter - 304 Stainless steel rod
4. SKU88 linear bearings
5. Mild steel rollers (machined using Turnmaster -45)
6. PLA 1.75mm diameter filament
7. MDF wooden board as base
8. Tin-lead solder
9. Micron precision screw gauge
10. Cuboidal geometry doctor's blade

VII. CONCLUSION



A compact, automated machine was successfully developed for the prototyping-scale fabrication of supercapacitor electrodes. The system integrates five essential processes—multi-stage sanding, cleaning, slurry coating, fan-assisted drying, and calendaring—into a continuous, modular workflow. The use of stepper motor-controlled substrate movement and adjustable doctor blade and calendaring modules ensures uniformity, reproducibility, and process control across electrode samples.

Characterization of the fabricated electrodes confirmed the effectiveness of the system. Surface roughness analysis demonstrated controlled reduction in R_a with progressive sanding, aiding slurry adhesion. Brookfield viscometry showed consistent slurry viscosity within an optimal processing range. A 12-hour soak test in 3M Na_2SO_4 electrolyte validated the mechanical stability and adhesion quality of coatings with higher binder content and smoother substrate preparation.

These results confirm that the developed machine provides a reliable platform for producing consistent electrode samples suitable for electrochemical testing. Future work will focus on automating additional steps, integrating drying temperature control, and coupling the system with real-time quality monitoring for scalable material screening and device prototyping.

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